

In-Vehicle Networking: Protocols, Challenges, and Solutions

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ABSTRACT

Fuel utilization efficiency and cost reduction are two major goals in designing in-vehicle networks. Aiming to address these two issues, we investigate in-vehicle networking protocols from both wired and wireless perspectives by first presenting representative solutions in each area, then identifying the challenges to current solutions, and finally advocating the use of the automotive Ethernet. Also, we propose a priority-based scheduler for the automotive Ethernet. Our preliminary experiments show that the proposed scheduler is effective and flexible, and thus is applicable to next-generation in-vehicle networks. We hope that further studies in this area can be inspired by our work and will be forthcoming in years to come.

INTRODUCTION

It is a norm now to find in modern cars a myriad of novel applications such as entertainment systems, navigation systems, on-board diagnostics, and augmented reality (AR) dashboards. Moreover, luxury vehicles are equipped with a variety of camera-based Advanced Driver Assistance Systems (ADASs) features including adaptive cruise control, lane shifting warning, automatic parking, and autonomous driving. These new applications are built upon wired in-vehicle electronics controlled by on-board electronic control units (ECUs). The constantly increasing number of ECUs installed in vehicles (some top-ranked automobiles already have approximately 70 ECUs on board [1]) presents new challenges in the sense that these ECUs demands broader in-vehicle network bandwidth and lower latency.

In addition to network performance issues, the wire harness itself of in-vehicle networking, following the engine and chassis [2], is the third most expensive system and the third heaviest in a vehicle. As such, there is a pressing need to simplify the current complex wiring in vehicles. New technologies that can reduce the weight of the harness system can directly result in better fuel efficiency. Also, with vehicles being designed smaller, the routing of the wire harness inside vehicles has become a thorny issue. To resolve these problems, new in-vehicle networking technologies are expected to be developed to reduce vehicles' weight and fuel consumptions.

Roughly speaking, an auto-electronics system consists of four subsystems: powertrain, chassis, body, and infotainment. Various protocols have been developed for these systems. For example,

the controller area network (CAN) has long been used to transmit the majority of in-vehicle communication signals and is still widely deployed in the powertrain and body control domains. FlexRay, with a distinguished determinism and fault-tolerance capability, is typically used in support of advanced chassis control and communication backbones. While the local interconnection network (LIN) was designed for cost-saving purposes and is often used in low-speed communications where high networking performance is usually not required, Media Oriented Systems Transport (MOST) networks are notably expensive and are commonly used in premium vehicles as the carrier of infotainment data.

Although substantial progress in in-vehicle networking has been made for auto-electronics, all of the aforementioned protocols have to be improved in terms of fuel efficiency and cost reduction. Aiming to address these two issues, in this article we investigate the in-vehicle networking protocols from the perspectives of both wired and wireless communications by first summarizing representative solutions in each area, then identifying challenges to the existing solutions, and finally prompting the use of automobile Ethernet. Also, a priority-based scheduler for automobile Ethernet is proposed. Our preliminary experiments show that the proposed scheduler is effective and flexible, and thus is applicable to next-generation in-vehicle networks.

AUTOMOBILE ELECTRONICS BASICS

Regarded as a system, automobile electronics can be divided into several sub-systems or domains, i.e., powertrain, chassis, body, and infotainment. Each sub-system has its own control units such as mechanical, electrical, or computer controls, which are independent of but cooperative with those of other sub-systems.

Powertrain: The powertrain sub-system refers to a set of automobile components including engine, transmission, shafts, wheels, and so on, that generate power to the vehicle. The powertrain can also include sensors and actuators to improve the comfort of the ride, reduce pollution caused by exhaust systems, increase fuel efficiency, and strengthen the vehicle's safety. Through these sensors and actuators, oil flow, cylinder pressure, speed, torque, angle, volume, position, and stability are measured and traced. Today's powertrains are even equipped with advanced computer-aided controls of power efficiency, and they typically require that precise results be obtained in microseconds.

Chassis: The chassis sub-system refers to the internal framework that supports the powertrain and all other components, except the engine, that are required for driving. Brakes, steering, and suspension are commonly known components in the chassis. Similar to the powertrain, sensors and actuators can be installed in the chassis domain and have stringent delay requirements.

Body: The body sub-system includes comfort-controlling components such as climate control, seat adjustment, window rolling, lights, and so on. Sensors for these components usually have low bandwidth requirements and have a relatively high-tolerance for delays (milliseconds).

Infotainment: The infotainment sub-system provides an interface for facilitating the interaction between humans and the automobile's electronics. This sub-system presents information acquired from sensors or ECUs in a user-friendly and interactive manner to users for the purpose of entertainment. Users' mobile devices can also be hooked up to the infotainment sub-system via Bluetooth, WiFi, or cellular networks. Moreover, as the infotainment sub-system controls different parts of the vehicle and displays their information, it is capable of communicating with other sub-systems. For example, some advanced infotainment systems can remotely diagnose vehicle problems by gathering diagnostic data from other sub-systems. As such, the infotainment system requires high bandwidth but is tolerant to delay in the scale of millisecond.

Summary of Automobile Electronics: A summary of the connectivity requirement for these sub-systems is shown in Table 1. Note that another sub-system that is vertical to these sub-systems is the driver assistance and safety system. This system is designed to assist drivers to operate a vehicle in a safer manner. It includes built-in GPS, cruise control, automatic parking, and even more advanced functionalities such as lane-shift warning, collision avoidance, intelligent speed adjustment or advice, driver drowsiness detection, and blind spot detection. This system typically has its own sensors and dedicated controllers, which intensively interact with other systems (i.e., powertrain, chassis, body, and infotainment). Also, it usually demands high bandwidth, and such a demand keeps increasing due to more cameras (for example, BMW's surround view feature) being installed on new car models.

IN-VEHICLE NETWORKING: WIRED INTERCONNECTION

Today's vehicles are equipped with more electronics than before to deal with the various needs of drivers and passengers and to improve fuel efficiency. A variety of sensors and actuators as well as ECUs are interconnected by multiple in-vehicle networking paradigms, exchanging information and coordinating controls to allow more useful functionalities to be implemented. In general, there are five representative in-vehicle networking technologies employed in the current automobiles: controller area network, local interconnect network, FlexRay, Media Oriented Systems Transport, and Ethernet. The state-of-the-art of these technologies is reviewed in this section.

Controller Area Network (CAN): CAN is an asynchronous serial bus network that interconnects devices, sensors, and actuators in a system

Sub-system	Description	Bandwidth requirement	Delay requirement	Reliability requirement
Powertrain	General and transmit power	Low	<10 μ s	High
Chassis	Control steering, brakes, and suspension	Low	<10 μ s	High
Body	Control A/C, windows, and lights	Low	<10 μ s	Medium
Infotainment	Display vehicle and multimedia info	High	<10 μ s	Medium

TABLE 1. Summary of connectivity requirements.

or a sub-system for control applications. CAN is a multi-master communication protocol that is designed for data integrity and automotive applications with data rates up to 1 Mb/s. In addition to the automotive applications, the CAN protocol is also used as a generic embedded communication system for microcontrollers and as an industrial control system. Over the last decade, two major physical-layer designs have been devised for the CAN protocol. Both communicate using a differential voltage on a pair of wires, and they are commonly referred to as high-speed and low-speed CAN. While high-speed CAN networks allow data transfer rates up to 1 Mb/s, low-speed/fault-tolerant CAN networks can communicate with devices at data rates up to 125 kb/s.

CAN is known for its low cost and high reliability. Due to these advantages, CAN is widely used in the powertrain, chassis, and body electronics. However, CAN has relatively low bandwidth and is a shared medium for data transmissions, which significantly restricts its application to other domains such as infotainment.

Local Interconnect Network (LIN): LIN is a universal asynchronous receiver-transmitter-based, single-master, multiple-slave networking protocol that was purposefully developed for automotive sensor and actuator networking applications. LIN offers a cost-effective alternative for connecting motors, switches, and sensors in the vehicle. The master node of LIN bridges the gap of the individual sensors and actuators by interconnecting them with higher-level networks, such as CAN.

Compared with CAN, LIN can be seen to have the following unique features. First, LIN facilitates effective communications for sensors and actuators where the bandwidth and versatility of CAN are not required. Second, LIN complements CAN as a cost-effective sub-network. Third, the LIN protocol can be generated by standard asynchronous communication interfaces, and thus no specific hardware is required. Fourth, there is no protocol license fee associated with LIN at all, as it is an open source protocol. As such, LIN is often used for body electronics as it is free and its bandwidth requirement is easy to meet.

FlexRay: FlexRay is a network communications protocol with a dual-channel data rate up to 10 Mb/s for advanced in-vehicle controls. The notable feature of FlexRay is its dual-channel architecture that provides reassurance to satisfy the reliability requirements of emerging safety systems such as brake-by-wire. The FlexRay protocol can also be employed as an in-vehicle network backbone,

Wireless communication technologies are a potential alternative for in-vehicle networking. These technologies not only allow wireless connections to be established between drivers/passengers' personal electronics and vehicles' built-in infotainment system, but can eliminate the need of wirelines by interconnecting sensors, actuators, and ECUs in such a manner as well.

working in conjunction with existing systems like CAN and LIN. FlexRay is a competitive alternative in terms of reducing costs in the sense that multiple CAN networks can be replaced by FlexRay in meeting bandwidth requirements in vehicles. It has been widely accepted that FlexRay can serve as the next-generation in-vehicle networking technology in enabling reliable management of various new safety and comfort features.

Unlike CAN, FlexRay leaves the task of correcting errors to its application layer, and thus has no error recovery mechanism. Also, with a much higher cost than CAN, FlexRay can be used in high-performance powertrain and safety applications, such as active suspension and adaptive cruise controls.

Media Oriented System Transport (MOST): MOST is a high-speed multimedia network technology that is specially designed for the infotainment system. Devices such as optical fibers or electrical conductors inside the car form a ring topology, and each device synchronously transmits audio and video signals to each other through the MOST protocol. Prominent features of MOST lie in its rate that can be as high as 50 Mb/s, and it is able to manage up to 64 MOST devices in a ring topology. Other topologies can also be configured with MOST; for example, safety-related applications typically use the double ring configurations.

Although MOST is capable of providing a relatively higher bandwidth than CAN, LIN, and FlexRay, its cost is, unfortunately, much higher than those protocols. Thus, MOST is only recommended for in-vehicle camera and/or video connections.

Ethernet: Ethernet has been the standard technology for local area networks ever since it was invented, and plays a critical role in the development of all types of communications. Automotive Ethernet is the Ethernet technology when it is used to connect components within a vehicle. Being initially designed to meet various (e.g., electric, bandwidth, latency, synchronization, and network management) requirements, the advantages of automotive Ethernet are obvious:

- It increases the communication bandwidth for advanced driving functionalities and the infotainment system.
- It changes the in-vehicle network structures from decentralized domain-specific topologies to hierarchical ones.
- It enhances the scalability and flexibility of future in-vehicle networking architecture.

In addition to these networking technologies, advancements in power and data transmissions are also worth noting.

Power Line Communication (PLC): PLC is a set of technologies using electronic wires to simultaneously carry both data and electric power. Traditionally, electronic devices in a vehicle were always required to have at least two connections:

one for data transmission and one for power supply. In the auto industry, conducting power and data transfers over the same line would be especially useful due to the resulting weight and cost reductions.

DC-BUS, developed by Yamar Electronics, is a proven competitive PLC technology for reliable and economical communications over noisy direct current (DC) power lines. For example, SIG60 and DCAN250 devices supported by the DC-BUS technology can operate as the power line communication physical layer transceivers for a CAN bus or a LIN bus. Another new Ethernet PLC solution, 1-Pair Power over Data Lines (PoDL), standardized by IEEE 802.3bu, is designed to provide sufficient power for devices over a single unshielded wire pair while carrying data at the same time. PoDL is capable of delivering power at 12V, 24V, and 48V DC levels.

IN-VEHICLE NETWORKING: WIRELESS INTERCONNECTION

Wireless communication technologies are a potential alternative for in-vehicle networking. These technologies not only allow wireless connections to be established between drivers/passengers' personal electronics and vehicles' built-in infotainment system, but can eliminate the need for wirelines by also interconnecting sensors, actuators, and ECUs in such a manner. To successfully consolidate wireless communication technologies in a vehicle, high bandwidth is commonly required in the infotainment subsystem to transmit multimedia content, and communications among devices/subsystems usually demand high reliability and low latency. Specifically, the unique features of wireless communications technologies make them suitable for body, powertrain, and chassis subsystems. In this section, several representative wireless technologies, which have been used in in-vehicle networking or could be potentially deployed in the future, are comprehensively summarized.

WiFi: Wireless Fidelity (WiFi), standardized by IEEE 802.11, is designed to replace high-speed cabling for general local area networks. In the in-vehicle networking environment, WiFi can be used to connect the infotainment subsystem and consumers' electronics, including smart phones, tablets and laptops. With a maximum data-rate of 600 Mb/s (IEEE 802.11n), WiFi offers abundant bandwidth to transmit multimedia data. When in-car WiFi meets cellular communication [3], its potential will be completely unlocked. With portable WiFi hotspots providing Internet access on the road, drivers can acquire real-time traffic status on GPS, and passengers can watch videos online while they are traveling.

Bluetooth: Bluetooth technology is designed for short-range wireless communications. Bluetooth BR/EDR (Bluetooth 2.1) with a maximum throughput of 3 Mb/s is optimized for audio streaming and is ideal for wireless speakers, headsets and hands-free in-car audio systems. Bluetooth-enabled hands-free calling systems are now available as standard equipment in over 90 percent of new vehicles sold today. Going beyond the hands-free calling feature, Bluetooth 2.1 can

Specification	Topology	Data rate	Channel bandwidth	Distance	Latency	TX power
IEEE 802.11n	Star	150–600 Mb/s	40 MHz	70 m (indoor)	10–100 ms	15–20 dBm
Bluetooth 2.1	Point-to-point	2–3 Mb/s	1 MHz	100 m	100+ ms	0–20 dBm
UWB	Point-to-point	53–480 Mb/s	≥ 500 MHz	3–10 m	Evolving	–41.3 dBm/MHz
Bluetooth LE	Point-to-point, Star, Mesh	305 kb/s	2 MHz	50 m	3 ms	–20–10 dBm
ZigBee	Point-to-point, Star, Mesh	20–250 kb/s	2 MHz	100 m	16 ms	–32–0 dBm

TABLE 2. Comparison of WiFi, Bluetooth and wireless USB [4–6].

also bring music to the car by connecting users' smartphones to the in-car audio system. However, unlike WiFi, Bluetooth 2.1 is suited for transmitting video streams.

Bluetooth LE (Bluetooth 4.0) enables connections with low duty cycle, low latency, low energy consumption and high reliability. It supports mesh topology for large-scale device networks like building automation, sensor networks and any scenario with multiple connected devices. Moreover, it has the potential to be deployed in the body comfort subsystem, powertrain subsystem, and chassis subsystem to interconnect ECUs, sensors and actuators. This makes possible the elimination of wires, thereby reducing weight, improving fuel efficiency, and lowering costs. However, it remains open as to whether wireless communication is safe for critical systems like the brake-by-wire or steer-by-wire system.

UWB: Ultra-wideband (UWB) is a technology developed for short-range, high-bandwidth communications, which transmits signals spread over a large bandwidth (≥ 500 MHz) at a very low energy level. The unlicensed use of UWB authorized by the FCC is in the frequency range from 3.1 GHz to 10.6 GHz. A typical UWB platform is capable of sending 480 Mb/s at a distance up to 3 meters and 110 Mb/s at up to 10 meters. Unlike spread spectrum communications, UWB transmits signals in a way that does not interfere with conventional narrowband and carrier wave transmissions in the same frequency band. UWB can be considered as a futuristic alternative to WiFi or Bluetooth.

ZigBee: ZigBee, built on the IEEE 802.15.4 standard, is a mature wireless interconnection solution that is designed to address the needs of low-cost, low-power wireless machine-to-machine networks. It utilizes unlicensed bands to provide wireless connections with data rates up to 250 kb/s. ZigBee supports multiple network topologies such as point-to-point, point-to-multipoint, and mesh networks. Devices in a ZigBee network are usually divided into two categories: reduced function device (RFDs) and full function device (FFDs). While FFDs are able to set up a mesh network with a low duty cycle, RFDs can join the mesh network as leaf nodes.

Typical ZigBee applications include wireless sensor networks, home automations, and industrial controls. Like the role played by Bluetooth LE in in-vehicle networks, ZigBee can be used to interconnect ECUs, sensors, and actuators in a car for the purposes of in-vehicle applications' controlling, monitoring, and status reading.

A comparison of WiFi, Bluetooth BR/EDR, UWB, and ZigBee is shown in Table 2.

Although some wireless communication technologies such as WiFi and wireless USB are able to provide high bandwidths, the interference caused by a roaring engine to wireless communications would be problematic, as it would compromise the performance of wireless communications in such a complicated electromagnetic environment and reduce the bandwidth.

CHALLENGES

Though the technologies summarized in the previous section have been successfully applied to in-vehicle networking, some challenges in in-vehicle networking remain to be addressed.

Bandwidth: With more and more "fancy" features being developed for a better driving and riding experience in the car, automotive electronics have become much more complicated. More sensors, actuators, ECUs, and micro-computers need be installed in future vehicles and the interactions between different sub-systems/domains will be highly intensive. This situation calls for a new in-vehicle networking technology with higher bandwidth support. From the discussion of present in-vehicle networking technologies, it can be seen clearly that the Ethernet, which is able to provide up to 1 Gb/s data rate, is an ideal solution for this issue. Moreover, unifying different protocols of various sub-systems/domains by Ethernet can lower the communication barrier for those systems, leading to easier and efficient interactions among them.

In the current automotive industry, BMW is the pioneer in employing Ethernet to address the bandwidth issue. Following BMW's lead, other car manufacturers have started implementing Ethernet technology in their automobiles. For instance, Hyundai and Volkswagen use Ethernet in their infotainment system and driver assistance system, respectively.

Note that although some wireless communication technologies such as WiFi and wireless USB are able to provide high bandwidths, the interference caused by a roaring engine to wireless communications would be problematic, as it would compromise the performance of wireless communications in such a complicated electromagnetic environment and reduce the bandwidth.

Latency: Apart from the bandwidth requirement, latency is another critical measurement in in-vehicle networking since most of the safety-related modules and powertrain as well as chassis controls in a vehicle all have strict delay requirements for the sake of reliability guarantee. Latency can be affected by various factors, especially when communications take place across different domains. For example, consider a data transmis-

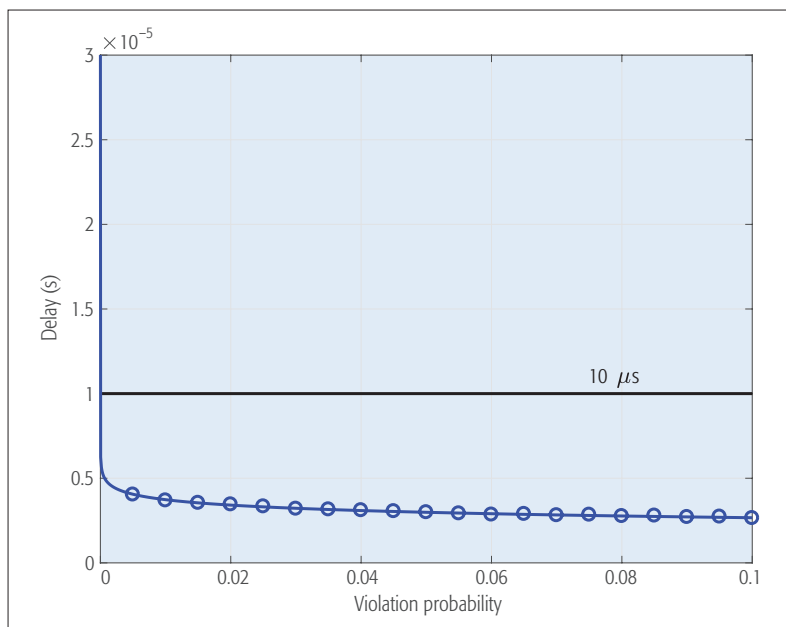


FIGURE 1. Delay bound of high-priority flow.

sion that is interfered by some noise signals. Such a transmission will probably have to be resent due to transmitting errors caused by the interference, which will inevitably incur delays as opposed to noise-free transmissions. Another example is the mismatch of delay constraints between two domains, say powertrain and infotainment. An instant fuel reading at the moment is obviously not the fuel consumption at that time (should be some previous time), and any control action from infotainment to the powertrain/chassis (e.g., air-bag on/off) will definitely experience a delay to take effect.

Although Ethernet technology is able to provide rich bandwidth for in-vehicle networking, its accessibility shared by media presents the challenging issue of the compliance of the delay constraints of the safety system and the vehicle controls. Therefore, new automotive Ethernet techniques are needed to meet the vehicle's latency requirements.

Likewise, even though Bluetooth LE is designed for low-latency applications, wireless communications would have a higher latency than wired communications. For some delay-sensitive applications in vehicles, wireless communications may not be a good option due to its inability to guarantee the desired latency requirements.

Cost: The multiple standards of wiring in vehicles and the various wired/wireless cabling have made the wire harness more complicated and costly than ever. Being done at one time, the wire harness accounts for 50 percent of the total labor in manufacturing vehicles. Moreover, the wire harness and the wire system are ranked the third most expensive and the third heaviest component in vehicles, respectively (with the powertrain being first and the chassis second in both cost and weight). As such, reducing the wire system weight can contribute to fuel efficiency directly [7].

In the past, LIN was considered as a solution for reducing the cost of in-vehicle networking since LIN nodes are relatively cheap (\$1.50 per node) and there is no license fee associated with

the LIN protocol. However, the bandwidth of LIN is extremely low, which might not be enough to support the applications in future vehicles. Ethernet, again, becomes a legitimate contender in this case, as it is cost-manageable to install the Ethernet interface for each network component and Ethernet can provide abundant resources for those bandwidth-needing applications such as autonomous driving.

Although wireless communication does not need copper wiring to transmit information, it does need wireless modules such as antennas and dedicated chips to be installed. It remains unclear whether replacing copper wiring with wireless communications in in-vehicle networking is more economical or not. Considering that most of the ECUs, sensors and actuators in the in-vehicle network are powered by an electrical line, power line communication should be a promising approach to reducing weight and cost. Furthermore, other wireless charging technologies such as simultaneous wireless information and power transfer (SWIPT) could even provide a wire-free charging solution for in-vehicle networking.

Security: The prevailing vehicular ad-hoc network (VANET) technologies, such as dedicated short range communications (DSRC), long term evolutionary vehicle (LTE-V) and WiMAX, present unprecedented challenges to the security and privacy of in-vehicle networking. Today's vehicles are expected to connect with one another wirelessly and even be connected to the Internet directly. As such, a vehicle is no longer an isolated system. In particular, not only will the vehicle suffer from the same security issues as in wireless networks, but it will be interfered by various surrounding noises as well [8]. This may lead to a greater threat to the safety of drivers and passengers. Also, like any computers connected to the Internet, connected vehicles can be attacked, hijacked, and jeopardized remotely by malicious users or even terrorists.

The issue of security is even more pressing in the context of wireless in-vehicle networking. Compared with wired communications, wireless communications can be easily eavesdropped and interfered. Rouf *et al.* [9] raised concerns about wireless in-vehicle networking security, demonstrating that eavesdropping on a tire pressure monitoring system, which uses wireless communication to transfer sensor data and even reverse engineering to inject false data, is likely to occur in a moving vehicle.

Current in-vehicle networking technologies, unfortunately, do not address these security and privacy issues. As the number of vehicles being wirelessly connected keeps growing, connected private vehicles, buses and public transportation facilities can eventually build an intelligent transportation system, which will soon make the in-vehicle networking security issue become a crucial problem that must be dealt with [10].

A PRIORITY-BASED SCHEDULER FOR THE AUTOMOTIVE ETHERNET

System Overview: Due to its scalable bandwidth, networking flexibility, and potential capability connecting to other vehicles and roadside infrastructures, Ethernet has been regarded as a

competitive in-vehicle networking technology for future automobiles. It is envisioned that the automotive wire harness structure will be changed from heterogeneous networks of various protocols to hierarchical and homogenous Ethernet networks. However, the following issues associated with Ethernet technology are notable [11]:

- It does not allow much RF “noise” and is vulnerable to the noise signals produced from other components in the vehicle.
- Due to its shared medium, it is unable to pinpoint latency down to the microsecond range, which is, however, required by any (fast) sensor/control reactions.
- It does not have a bandwidth allocation scheme allocating individually needed bandwidths to different data streams, resulting in the speculation that it might not be suitable to efficiently transmit shared data from multiple types of sources.
- It does not have any mechanisms for synchronizing devices and having multiple devices sample data at the same time.

To address the latency issue mentioned above, a priority-based scheduler is proposed in this article for automotive Ethernet networks. In the scheduler, three queues with high, medium, and low priorities are set up for powertrain/chassis control flow, body control flow, and infotainment data stream, respectively. The high-priority flow is scheduled preemptively over the other two flows. Evidently, the powertrain/chassis control flow in our scheduler has a strictly high priority. As such, the latency requirement of each type of flows is expected to be guaranteed to be met in the vehicle.

Simulation Results: We denote high-, medium-, and low-priority flows as $flow_1$, $flow_2$, and $flow_3$, respectively, which follow the *Gaussian* traffic arrival model described in [12]. The scheduler can be regarded as a server providing accessibility to the shared medium [13]. A latency-rate service model is employed with constant service rate C and latency $(L^{ma})/C$. Note that the server is shared by three flows under a strict priority scheduling. Stochastic network calculus is leveraged to analyze the latency performance of the proposed scheduler based on the service model and the arrival model [14]. Stochastic network calculus is able to provide a probabilistic bound, describing the upper bound on the probability of delay exceeding a certain maximum level. In other words, such a probabilistic bound can guarantee a certain upper bound on delay with a maximum violation probability.

The detailed parameter settings are as follows: $\hat{r}_1 = \hat{r}_2 = \hat{r}_3 = 0.5$ Mb/s; $\hat{v}_1 = \hat{v}_2 = \hat{v}_3 = 0.75$; $C = 1000$ Mb/s; $L^{max} = 200$ Bytes, $r_1 = 900$ Mb/s, $r_2 = 30$ Mb/s, $r_3 = 30$ Mb/s. For the meaning of each parameter we refer readers to [15].

Figure 1 shows the stochastic delay bound of the high-priority flow with respect to the violation probability. As can be seen from this figure, the delay of $flow_1$ decreases and stays constantly below $10 \mu s$ (solid horizontal line) as the violation probability increases. For example, to the point (0.1, 2.5) in this curve, it means at least 90 percent of the time the delay level would not higher than 2.5 ms. Moreover, the intersection of the two lines is extremely close to the point where the violation probability equals zero. These observations clearly indicate that the $10 \mu s$ delay requirement

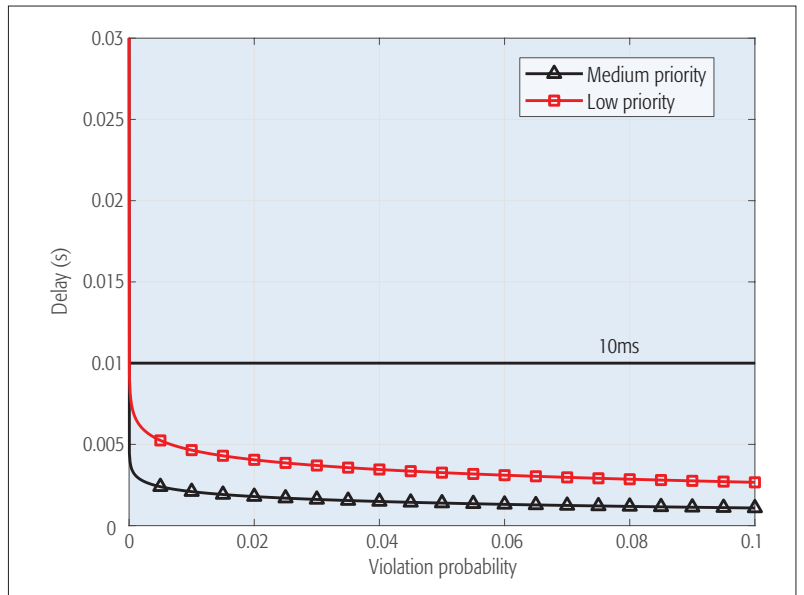


FIGURE 2. Delay bounds of medium- and low-priority flows.

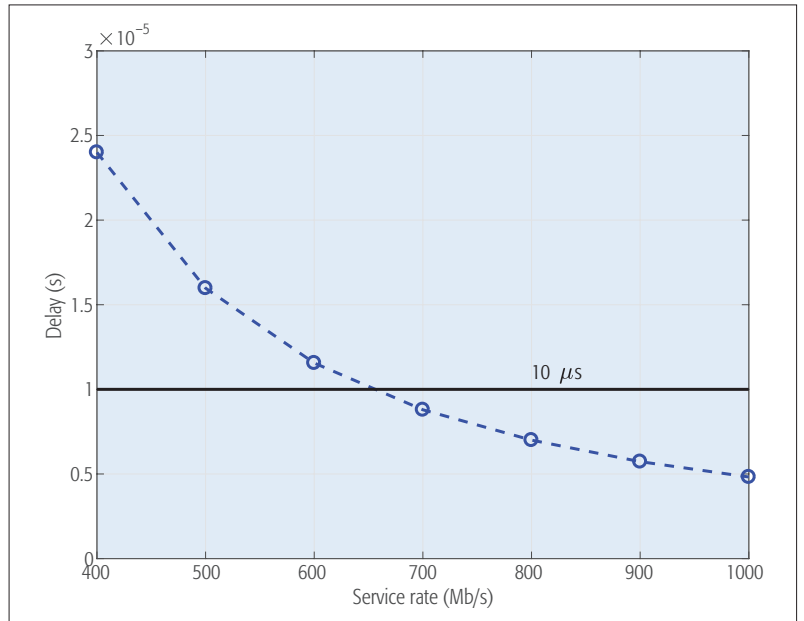


FIGURE 3. Correlation between service rate and delay bound of high-priority flow with violation probability 0.001.

(Table 1) for the high-priority flow (powertrain and chassis control messages) is met by our scheduler. In a similar fashion, Fig. 2 demonstrates that the medium- and low-priority flows (body and infotainment data transmissions) can also meet their delay requirements (10 ms , solid horizontal line) with the scheduler. Therefore, our priority-based scheduler is effectively designed.

With a small violation probability, we also investigate the impact of service rate C on delays. Figures 3 and 4 respectively show the correlations between the service rate C and the delay bound of $flow_1$, $flow_2$, and $flow_3$ with violation probability being 0.001. It is evident that the delay bounds of the three flows are fully capable of meeting their respective delay requirements. This result further confirms the fact that our priority-based scheduler is effectively designed, and indicates that the scheduler can be applied to Ethernet-ori-

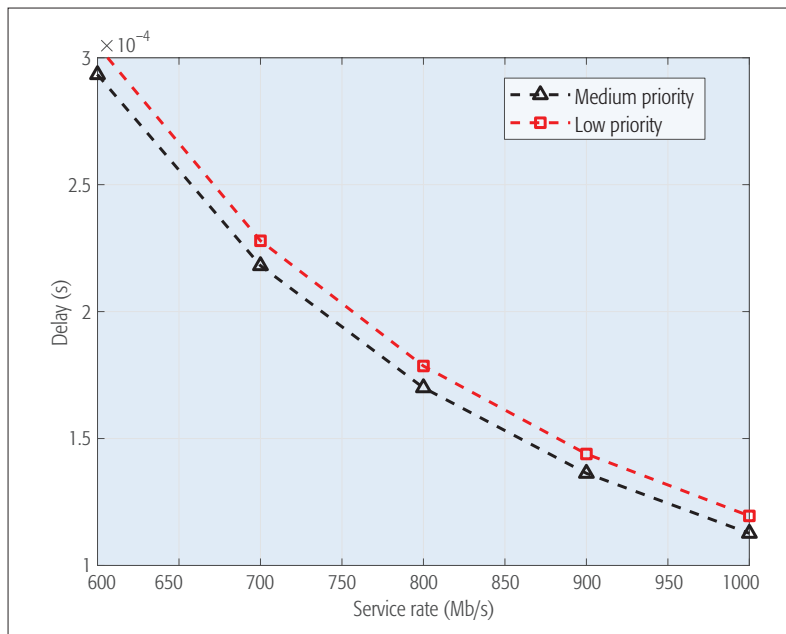


FIGURE 4. Correlations between service rate and delay bounds of medium- and low-priority flows with violation probability 0.001.

ented in-vehicle networks. Also, one may note in Fig. 4 that the difference between medium and low priority is not significant, because the delay level is more sensitive to the variation of service rate, and thus the gap caused by the priority scheduling is small. However, scheduling policy still matters in this case for a given service rate.

CONCLUSION

In this article, we have taxonomically summarized the in-vehicle networking protocols from both the wired and wireless perspectives. The challenges to the current practice have been identified. Aiming to improve fuel efficiency and reduce the cost of in-vehicle networking, we have advocated the automotive Ethernet and have proposed a priority-based scheduler for it. Both analyses and simulation results suggest that the proposed scheduler is effective and flexible, and is suitable for next-generation in-vehicle networks.

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